

The Program in Action

When you run *ninetyninebottles2.py*, the output will look like this:

```
niNety-nniinE Bo0ttels, by Al Sweigart al@inventwithpython.com
--snip--
99 bottles of milk on the wall,
99 bottles of milk,
Take one down, pass it around,
98 bottles of milk on the wall!

98 bottles of milk on the wall,
98 bottles of milk,
Take one d wn, pass it around,
97 bottles of milk on the wall!

97 bottles of milk on the wall,
97 bottels of milk,
Take one d wn, pass it around,
96 bottles of milk on the wall!
--snip--
75b otlte of mIl on teh wall,
75 ottels f milk,
Take one d wn, pass it ar und,
74 bb0ttles of milk on t e wall!
--snip--
1 otlE t of iml oo nteh lall,
1 o Tle FF FmMLIIkk,
Taake on d wn, pAasSs itt au nn d,
No more bottles of milk on the wall!
```

How It Works

String values in Python are *immutable*, meaning they cannot be changed. If the string 'Hello' is stored in a variable called `greeting`, the code `greeting = greeting + ' world!'` doesn't actually change the 'Hello' string. Rather, it creates a new string, 'Hello world!', to replace the 'Hello' string in `greeting`. The technical reasons for this are beyond the scope of this book, but it's important to understand the distinction, because it means code like `greeting[0] = 'h'` isn't allowed, since strings are immutable. However, since lists are mutable, we can create a list of single-character strings (as line 62 does), change the characters in the list, and then create a string from the list (line 85). This is how our program seemingly changes, or *mutates*, the strings containing the song lyrics.

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1. `"""niNety-nniinE Bo0ttels of Mlik On teh wall`
 2. `By Al Sweigart al@inventwithpython.com`
 3. `Print the full lyrics to one of the longest songs ever! The song`
 4. `gets sillier and sillier with each verse. Press Ctrl-C to stop.`
 5. `View this code at https://nostarch.com/big-book-small-python-projects`
 6. `Tags: short, scrolling, word"""`
 - 7.